CS 289A Fall 2026

Grad Project Outline

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This document is the shared outline for the final report. Sections and subsections below mirror the progress report's structure and the proposal's evaluation plan. Notes in comments are for us to fill in as the work progresses.

Introduction

Related Work

Post-Training of Foundation Models

Unsupervised Post-Training

Self-Consistency and Pseudo-Labeling

Visual Question Answering

Data

ScienceQA

VQAv2 (Stretch Goal)

Dataset Splits and Preprocessing

Methods

SmolVLM-500M-Instruct

Self-Consistency Sampling

Sampling Parameters

Mode and Frequency Extraction

Consistency Filter

Softmax-over-Options Variant

Read Position

Probability Normalization

LoRA Fine-Tuning

Experiments

Baselines

Zero-Shot Greedy

Corrected Softmax

Sampled-Mode ($k = 8$)

Oracle and $\tau = 0$ Fine-Tuning Comparisons

Ablations

Error-Propagation Study

Results and Discussion

Baseline Results

Post-Fine-Tuning Results

Against Greedy Baseline

Against Oracle and $\tau = 0$

Pseudo-Label Quality

Error Propagation

Limitations and Threats to Validity

Conclusion

References